APPLICATION OF SMART POWER SAVING DEVICE IN HOSTELS

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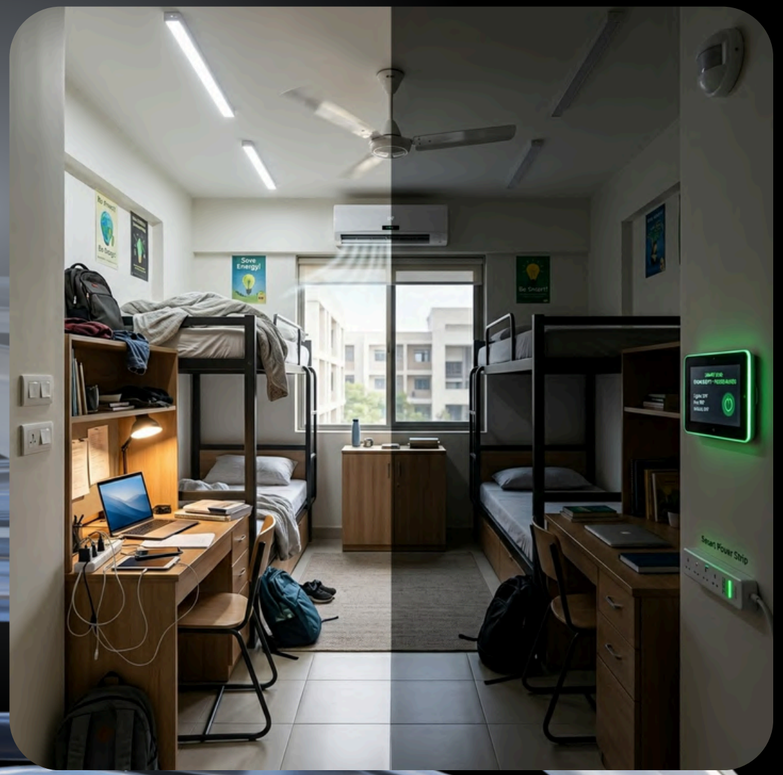


INTRODUCTION

- Electricity is one of the most important resources in hostels.
- Many students waste electricity by leaving fans, lights, and chargers ON.
- This causes higher electricity bills and energy loss.
- Smart power saving devices help reduce wastage automatically.

WHAT IS A SMART POWER SAVING DEVICE?

- A smart power saving device is an electronic system that controls power usage intelligently.
- It detects room occupancy and automatically switches appliances ON/OFF.
- It can be controlled manually or through mobile apps.

Examples:

- Smart switches
- Motion sensor lights
- Smart plugs
- Energy monitoring systems

NEED IN HOSTELS

- Large number of rooms consume high electricity.
- Students forget to switch OFF devices.
- Common areas use unnecessary lighting.
- Reduces maintenance cost.
- Promotes eco-friendly environment.

MAJOR POWER WASTAGE AREAS IN HOSTELS

- Room lights left ON during class hours.
- Fans running in empty rooms.
- Chargers plugged in continuously.
- Corridor lights ON all day.
- Water heater misuse.

OBJECTIVES OF THE SYSTEM

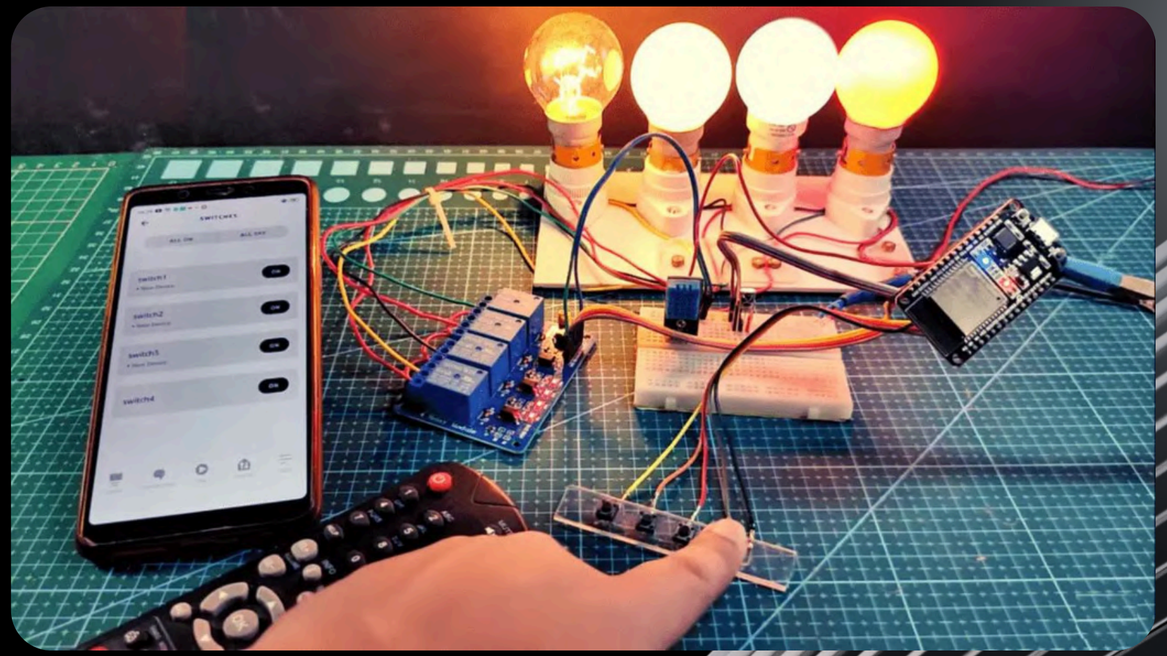
- Save electrical energy.
- Reduce monthly electricity bill.
- Automate room appliance control.
- Improve safety.
- Create awareness among students.

COMPONENTS USED

- Microcontroller (Arduino / ESP32)
- PIR Motion Sensor
- Relay Module
- Smart Meter
- Wi-Fi Module
- Mobile Application
- Temperature / Light Sensors

WORKING PRINCIPLE

- Sensor detects human presence.
- Controller checks room occupancy.
- Lights/fans turn ON automatically.
- If no motion detected for set time:
- Appliances turn OFF.
- Data sent to mobile app/admin panel.



BLOCK DIAGRAM

INPUT: MOTION SENSOR, LIGHT SENSOR

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CONTROLLER: ARDUINO / ESP32

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OUTPUT: RELAY CONTROL

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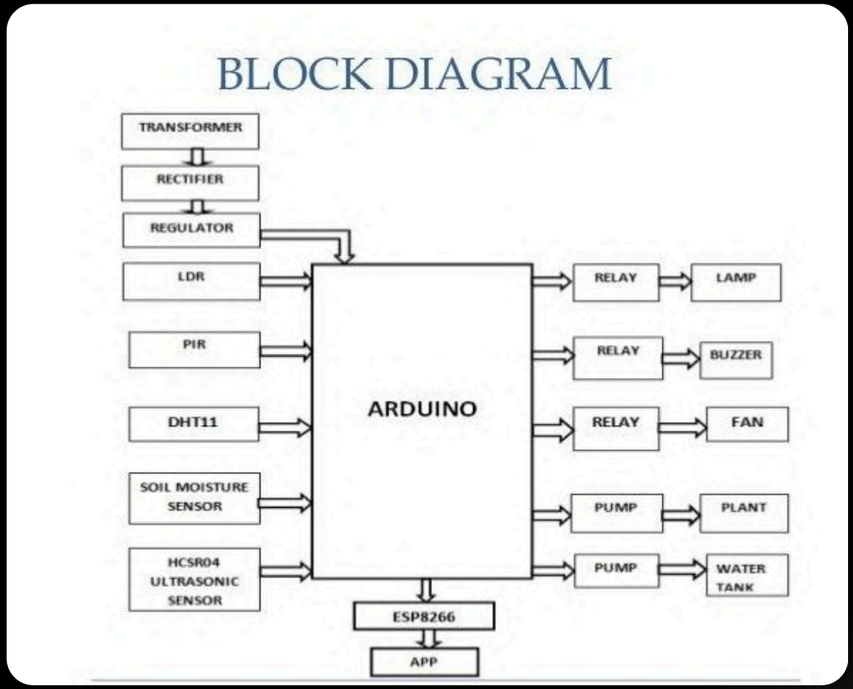
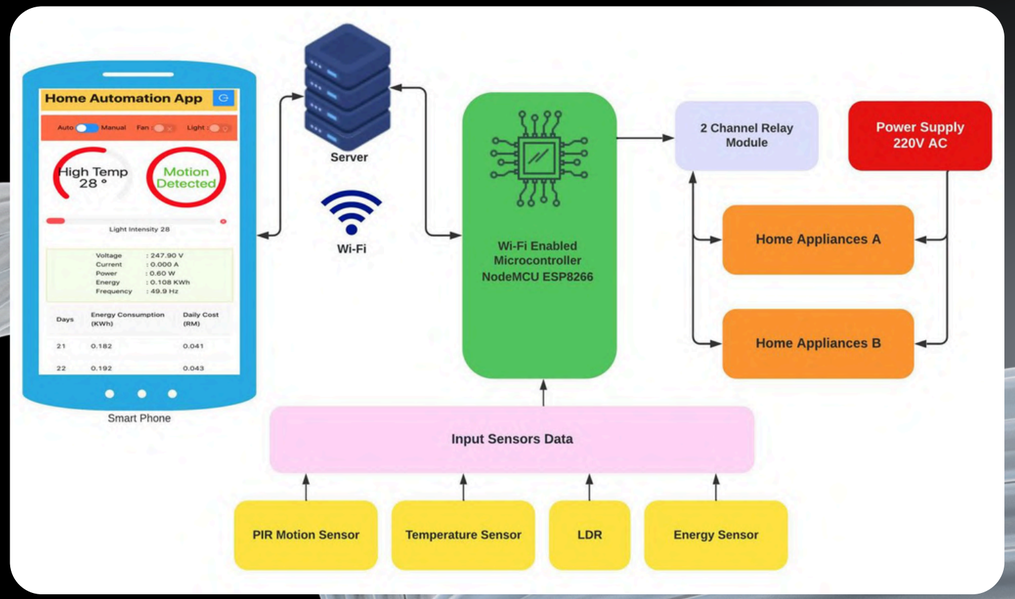
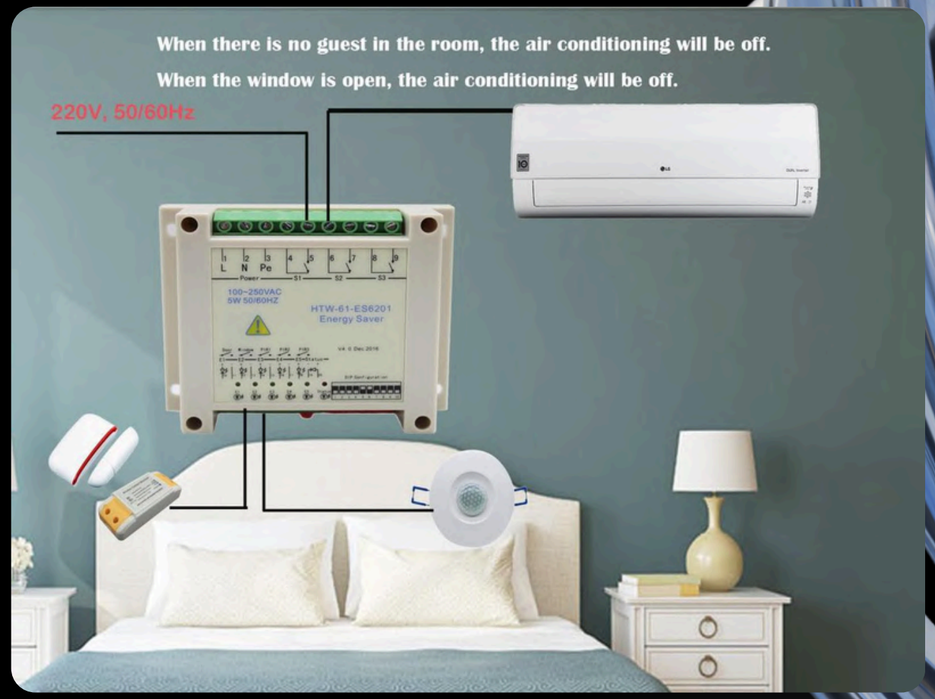
APPLIANCES: FAN, LIGHT, PLUG POINTS





SMART DEVICE IN HOSTEL ROOMS

- Auto OFF lights when students leave room.
- Fan speed control based on temperature.
- Power cut for idle chargers.
- Real-time energy monitoring.

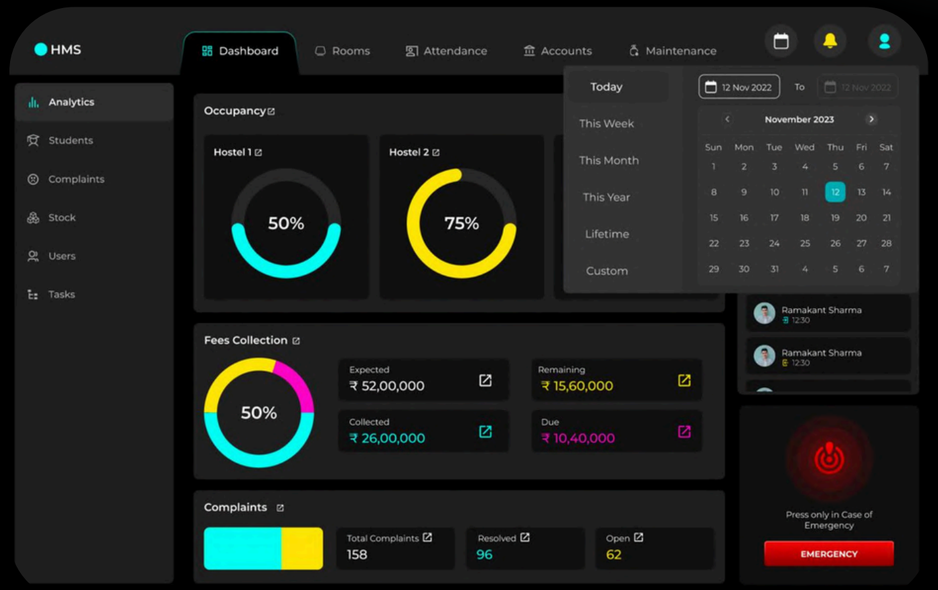


APPLICATION IN COMMON AREAS

- Staircase lighting using motion sensors.
- Auto corridor lights at night.
- Bathroom exhaust fan automation.
- Smart control in study hall.

MOBILE APP INTEGRATION

- Room-wise energy usage data.
- Remote appliance control.
- Monthly reports.
- Alerts for excess usage.



ADVANTAGES

- Saves 20% to 40% electricity.
- Lower hostel operating cost.
- Reduces human negligence.
- Better appliance life.
- Easy monitoring.

CASE STUDY EXAMPLE

HOSTEL WITH 100 ROOMS:

- Each room wastes 2 units/day.
- Total wastage = 200 units/day.
- Monthly wastage = 6000 units.

USING SMART DEVICE CAN SAVE MAJOR PORTION OF THIS ENERGY.

COST ESTIMATION

COMPONENT	COST
PIR Sensor	₹150
Relay	₹200
ESP32	₹500
Wiring	₹300
Total per room	₹1150

CHALLENGES

- Initial installation cost.
- Maintenance of sensors.
- Internet dependency.
- Student misuse or tampering.
- Need regular monitoring.

FUTURE ENHANCEMENTS

- AI based energy prediction.
- Face recognition room access.
- Solar power integration.
- Voice control system.
- Cloud monitoring dashboard.

ENVIRONMENTAL IMPACT

- Less electricity generation demand.
- Reduced carbon emissions.
- Supports green campus initiative.
- Sustainable hostel management.

CONCLUSION

Smart power saving devices are highly useful in hostels.

Improve automation and comfort.

They reduce energy waste and cost.

A smart hostel is the future of campus living.

THANK YOU